

可加载十进制计数器设计

1、实验任务

可加载数计数器输出四位二进制连接七段译码器模块，累加到 9 循环计数。

2、实验思路：

本实验利用 4 位二进制计数方法，通过时钟 CLK 上升沿触发一次，计数一次，从‘0000’一直计到‘1111’，再接“实验一”七段译码模块，通过累加到‘9’进行循环计数方法。其原理如图 1.1，利用一个四位二进制计数器，随着时钟上升沿到来，输出一组信号进行累加输出给七段译码器，译码器模块根据接受到的四位二进制信号进行对应译码点亮数码管七段。

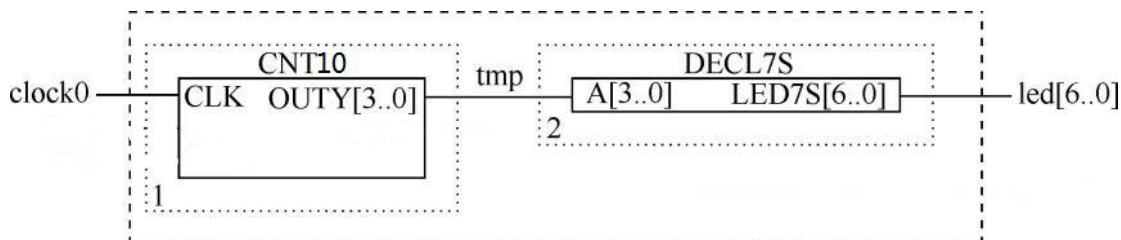


图 1.1

3、实验程序：

顶层文件：

```
module cnt10_disp
(
    input CLOCK,
    input [2:0] KEY,
    input [3:0] SW,
    output [7:0] SEL,
    output [7:0] DIG
);
localparam LEN = 25;
wire en,load_n,reset_n;
wire [3:0] d;
wire cout;
wire clk;
reg [6:0] LED7S;
reg [3:0] cnt;
```

```

reg [LEN-1:0] div;
assign {en,load_n,reset_n} = KEY;
assign DIG = {~cout,LED7S};
assign d = SW;
assign clk = div[LEN-1];
assign SEL = 8'b1111_1110;
always @(posedge clk,negedge reset_n)
    if(!reset_n)
        cnt <= 0;
    else if(en)
        if(!load_n)
            cnt <= d;
        else if(cnt < 9)
            cnt <= cnt + 1'b1;
        else
            cnt <= 0;
assign cout = (cnt==9);
always @*
    case(cnt)
        4'b0000 :   LED7S <= 7'b1000_000;//7'b0111_111;
        4'b0001 :   LED7S <= 7'b1111_001;//7'b0000_110;
        4'b0010 :   LED7S <= 7'b0100_100;//7'b1011_011;
        4'b0011 :   LED7S <= 7'b0110_000;//7'b1001_111;
        4'b0100 :   LED7S <= 7'b0011_001;//7'b1100_110;
        4'b0101 :   LED7S <= 7'b0010_010;//7'b1101_101;
        4'b0110 :   LED7S <= 7'b0000_010;//7'b1111_101;
        4'b0111 :   LED7S <= 7'b1111_000;//7'b0000_111;
        4'b1000 :   LED7S <= 7'b0000_000;//7'b1111_111;
        4'b1001 :   LED7S <= 7'b0010_000;//7'b1101_111;
        4'b1010 :   LED7S <= 7'b0001_000;//7'b1110_111;
        4'b1011 :   LED7S <= 7'b0000_011;//7'b1111_100;
        4'b1100 :   LED7S <= 7'b1000_110;//7'b0111_001;
        4'b1101 :   LED7S <= 7'b0100_001;//7'b1011_110;
        4'b1110 :   LED7S <= 7'b0000_110;//7'b1111_001;
        4'b1111 :   LED7S <= 7'b0001_110;//7'b1110_001;
        default :   LED7S <= 7'b1000_000;//7'b0111_111;
    endcase
always @(posedge CLOCK,negedge reset_n)

```

```

if(!reset_n)
    div <= 0;
else
    div <= div + 1'b1;
endmodule

```

4、实验仿真：

根据描述语言进行仿真测试，分析波形 图 1.2

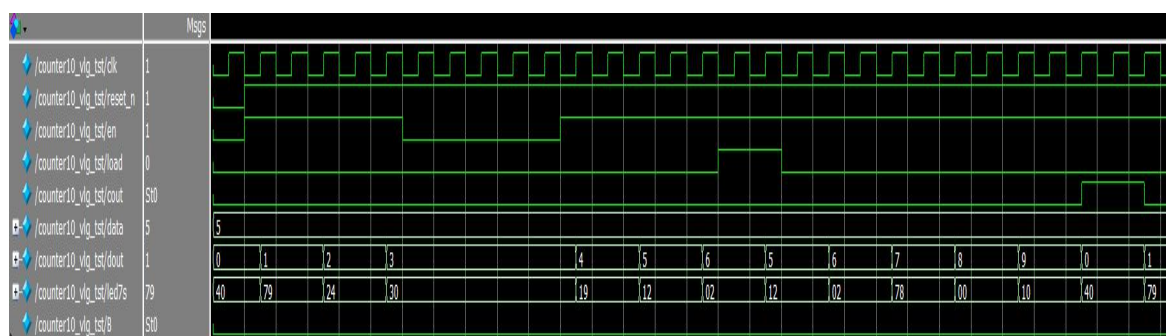


图 1.2 仿真结果

5、引脚约束：

复位、加载、使能分别接键、2、3，待加载数据分别接开关 SW1-4，拨上位 1，DIG/LED 开关拨下。

```

set_pin_assignment { CLOCK } { LOCATION = R7; IOSTANDARD = LVCMOS33;
PULLTYPE = PULLUP; }
set_pin_assignment { DIG[0] } { LOCATION = C15; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[1] } { LOCATION = C16; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[2] } { LOCATION = B6; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[3] } { LOCATION = A5; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[4] } { LOCATION = B5; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[5] } { LOCATION = A8; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }
set_pin_assignment { DIG[6] } { LOCATION = A7; IOSTANDARD = LVCMOS33;
DRIVESTRENGTH = 8; PULLTYPE = NONE; }

```

```
set_pin_assignment { DIG[7] } { LOCATION = A6; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { KEY[0] } { LOCATION = A2; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }  
set_pin_assignment { KEY[1] } { LOCATION = B2; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }  
set_pin_assignment { KEY[2] } { LOCATION = B1; IOSTANDARD = LVCMOS33;  
PULLTYPE = PULLUP; }  
set_pin_assignment { SEL[0] } { LOCATION = D11; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[1] } { LOCATION = E11; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[2] } { LOCATION = F10; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[3] } { LOCATION = C13; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[4] } { LOCATION = E10; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[5] } { LOCATION = E12; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[6] } { LOCATION = E16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SEL[7] } { LOCATION = F16; IOSTANDARD = LVCMOS33;  
DRIVESTRENGTH = 8; PULLTYPE = NONE; }  
set_pin_assignment { SW[0] } { LOCATION = C8; IOSTANDARD = LVCMOS33; PULLTYPE  
= PULLUP; }  
set_pin_assignment { SW[1] } { LOCATION = C7; IOSTANDARD = LVCMOS33; PULLTYPE  
= PULLUP; }  
set_pin_assignment { SW[2] } { LOCATION = C6; IOSTANDARD = LVCMOS33; PULLTYPE  
= PULLUP; }  
set_pin_assignment { SW[3] } { LOCATION = C5; IOSTANDARD = LVCMOS33; PULLTYPE  
= PULLUP; }
```

6、下载测试:

开关 SW1-SW3 输入待加载数，键 1 复位，按下去复位，松开计数，键 2 加载输入开关输入的数据，键 3 使能，按下去停止计数，松开使能。

